Recitation 4.

Problem 1. Let $X_1, \ldots, X_n$ be i.i.d. from the density

$$f(x|\theta) = \frac{x}{\theta^2} e^{-x^2/(2\theta^2)}, \ x \geq 0.$$

(1) Find the method of moments estimator of θ.

(2) Find the maximum likelihood estimator of θ.

(3) Find the asymptotic variance of the mle.

Problem 2. Suppose $X = (X_1, X_2, X_3) = (10, 68, 112)$ is an observation from a multinomial distribution with parameters $n = 190$ and $p_1 = (1-\theta)^2$, $p_2 = 2\theta(1-\theta)$ and $p_3 = \theta^2$ for some $0 < \theta < 1$.

(1) Compute the maximum likelihood estimator $\widehat{\theta}$ of θ, and its standard error.

(2) Compute an approximate 95% confidence interval for θ.

Problem 3. Let $X_1, \ldots, X_n$ be i.i.d. Exponential$(1/\theta)$.

(1) Find the MLE of θ.

(2) Show that $2X_1/\theta$ has an Exponential$(1/2)$ distribution. Deduce the distribution of $2\sum_{i=1}^n X_i/\theta$.

(3) Find the mean and variance of the MLE.

(4) Find the form of an approximate confidence interval for θ.

(5) Find the form of an exact confidence interval for θ. Hint: use part (2).

Solutions

Problem 1. Let $X_1, \ldots, X_n$ be i.i.d. from the density

$$f(x|\theta) = \frac{x}{\theta^2} e^{-x^2/(2\theta^2)}, \ x \geq 0.$$

(1) Find the method of moments estimator of θ.
Note that

$$\mu_1 = \mathbb{E}[X_1] = \int_0^\infty \frac{x^2}{\theta^2} e^{-\frac{x^2}{2\theta^2}} dx = \frac{1}{2}\frac{\sqrt{2\pi}}{\theta}\theta^2 = \sqrt{\frac{\pi}{2}}\theta \qquad \Longleftrightarrow \qquad \theta = \sqrt{\frac{2}{\pi}}\mu_1.$$

Hence

$$\widehat{\theta}_{MOM} = \sqrt{\frac{2}{\pi}}\overline{X}.$$

(2) Find the maximum likelihood estimator of θ.
First, we write out the likelihood function, log-likelihood function and its derivative

$$\begin{aligned} lik(\theta) &= \prod_{i=1}^n f(x_i|\theta) = \frac{1}{\theta^{2n}} \prod_{i=1}^n x_i e^{-\frac{\sum_{i=1}^n x_i^2}{2\theta^2}}; \\ \ell(\theta) &= \sum_{i=1}^n \log x_i - 2n\log\theta - \sum_{i=1}^n \frac{x_i^2}{2\theta^2}; \\ \ell'(\theta) &= -2n\theta^{-1} + \theta^{-3}\sum_{i=1}^n x_i^2. \end{aligned}$$

Setting $\ell'(\theta)$ to zero yields

$$-2n\theta^{-1} + \theta^{-3}\sum_{i=1}^n x_i^2 = 0 \Longrightarrow \widehat{\theta}_{MLE} = \sqrt{\frac{\sum_{i=1}^n X_i^2}{2n}}.$$

(3) Find the asymptotic variance of the mle.
First, we find the second derivative of the log density function

$$\frac{d^2}{d\theta^2}\log f(x|\theta) = 2\theta^{-2} - 3x^2\theta^{-4}.$$

Second,

$$\begin{aligned}\mathbb{E}\left[X^2\right] &= \int_0^\infty \frac{x^3}{\theta^2} e^{-\frac{x^2}{2\theta^2}} \mathrm{d}x \\ &= -\int_0^\infty x^2 \mathrm{d}e^{-x^2/(2\theta^2)} \\ &= -\int_0^\infty y \mathrm{d}e^{-y/(2\theta^2)} \\ &= \int_0^\infty e^{-y/(2\theta^2)} \mathrm{d}y = 2\theta^2.\end{aligned}$$

Hence, the Fisher Information is

$$I(\theta) = -\mathbb{E}\left[\frac{d}{d\theta^2} \log f(X_1|\theta)\right] = 3\theta^{-3}\mathbb{E}\left[X^2\right] - 2\theta^{-2} = 4\theta^{-2}$$

and the asymptiotic variance of the MLE $\widehat{\theta}_{MLE}$ *is* $1/\{nI(\theta)\} = \theta^2/(4n)$.

Problem 2. Suppose $X = (X_1, X_2, X_3) = (10, 68, 112)$ is an observation from a multinomial distribution with parameters $n = 190$ and $p_1 = (1-\theta)^2$, $p_2 = 2\theta(1-\theta)$ and $p_3 = \theta^2$ for some $0 < \theta < 1$.

(1) Compute the maximum likelihood estimator $\widehat{\theta}$ of θ, and its standard error.
Recall

$$\widehat{\theta}_{MLE} = \frac{2X_3 + X_2}{2n} = \frac{292}{380} \simeq 0.7684$$

and

$$\sqrt{Var(\widehat{\theta}_{MLE})} = \sqrt{\frac{(1-\theta)\,\theta}{2n}} \simeq \sqrt{0.000466} \simeq 0.0216.$$

`There is no need to rederive the formulas that we obtained in class!`

(2) Compute an approximate 95% confidence interval for θ.
Using the standard error we got from previous part, a 95% confidence interval for θ *is*

$$\begin{aligned}&(0.7684211 - 0.02158703 \times 1.96, 0.7684211 + 0.02158703 \times 1.96) \\ =&\,(0.7261105, 0.8107317)\,.\end{aligned}$$

Problem 3. Let $X_1, \ldots, X_n$ be i.i.d. Exponential$(1/\theta)$.

(1) Find the MLE of θ.
Observe that

$$lik(\theta) = \prod_{i=1}^{n} f(x_i|\theta) = \frac{1}{\theta^n} e^{-\frac{1}{\theta}\sum_{i=1}^{n} x_i},$$

$$\ell(\theta) = -n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i,$$

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i.$$

Setting $\ell'(\theta)$ to zero yields

$$\frac{1}{\theta^2}\sum_{i=1}^{n} x_i - \frac{n}{\theta} = 0 \Longrightarrow \widehat{\theta} = \frac{\sum_{i=1}^{n} X_i}{n} = \overline{X}.$$

(2) Show that $2X_1/\theta$ has an Exponential$(1/2)$ distribution. Deduce the distribution of $2\sum_{i=1}^{n} X_i/\theta$.
Let $Y = 2X_1/\theta$.

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\left(\frac{2X_1}{\theta} \le y\right) = \mathbb{P}\left(X_1 \le \frac{\theta y}{2}\right)$$
$$= 1 - e^{-\frac{1}{\theta}\cdot\frac{\theta y}{2}} = 1 - e^{-\frac{y}{2}}.$$

Hence

$$f_Y(y) = \frac{1}{2} e^{-\frac{1}{2}y},$$

which is the density of an exponential distribution with parameter $\frac{1}{2}$, that is, $2X_1/\theta \sim Exp(1/2)$. It follows that $2\sum_{i=1}^{n} X_i/\theta \sim Gamma(n, 1/2) = \chi^2_{2n}$.

(3) Find the mean and variance of the MLE. (Exact calculation required).
Note that $Y = 2X/\theta$ with $2 = \mathbb{E}[Y] = 2\theta^{-1}\mathbb{E}[X]$ and $4 = Var(Y) = 4\theta^{-2}\,Var(X)$, whence $\mathbb{E}[X] = \theta$ and $Var(X) = \theta^2$.

$$\mathbb{E}[\widehat{\theta}_{MLE}] = \mathbb{E}[\overline{X}] = \mathbb{E}[X] = \theta$$

$$Var[\widehat{\theta}_{MLE}] = Var[\overline{X}] = \frac{\theta^2}{n}.$$

(4) Find the form of an approximate confidence interval for θ.
We compute the asymptotic variance $1/\{nI(\theta)\} = \theta^2/n$ *as*

$$\frac{d^2}{d\theta^2} \log f(x|\theta) = \frac{1}{\theta^2} - \frac{2x}{\theta^3},$$

whence

$$I(\theta) = -\mathbb{E}\left[\frac{1}{\theta^2} - \frac{2X}{\theta^3}\right] = \frac{1}{\theta^2}.$$

The approximate $1-\alpha$ *confidence interval*

$$\left(\widehat{\theta}_{MLE} \pm z\left(\frac{\alpha}{2}\right) \frac{1}{\sqrt{nI\left(\widehat{\theta}_{MLE}\right)}}\right) = \left(\overline{X} \pm z\left(\frac{\alpha}{2}\right) \frac{\overline{X}}{\sqrt{n}}\right)$$

(5) Find the form of an exact confidence interval for θ. *Hint: use part (2).*
By Part (2), we see that $2n\overline{X}/\theta \sim \chi^2_{2n}$. *Let* $k_n(\alpha)$ *be the* α *upper point of* χ^2_n. *Then*

$$\mathbb{P}\left(k_{2n}(1-\alpha/2) \leq \frac{2n\overline{X}}{\theta} \leq k_{2n}(\alpha/2)\right) = 1-\alpha.$$

A simple algebra yields the exact $(1-\alpha)\,\%$ *confidence interval*

$$\left(\frac{2n\overline{X}}{k_{2n}(\alpha/2)},\ \frac{2n\overline{X}}{k_{2n}(1-\alpha/2)}\right)$$

`Not required: Note that many solutions are possible, any pair` (a,b) `with`

$$\mathbb{P}\{a \leq \chi^2_{2n} \leq b) = 1-\alpha$$

yield an exact $1-\alpha$ confidence interval

$$\left(\frac{2n\overline{X}}{b},\ \frac{2n\overline{X}}{a}\right)$$

for θ.